

Xi'an Jiaotong-Liverpool University

西交利物浦大學

PAPER CODE	EXAMINER	DEPARTMENT	TEL
INT202		INTELLIGENT SCIENCE	

2nd SEMESTER 2024/25 RESIT EXAMINATION

UNDERGRADUATE – YEAR 3

COMPLEXITY OF ALGORITHMS

TIME ALLOWED: 2 HOURS

INSTRUCTIONS TO CANDIDATES

1. This is a closed-book examination.
2. Total marks available are 100. This exam will count for 100% of the final mark.
3. Answer all questions. There is NO penalty for providing a wrong answer.
4. Only English solutions are accepted. Answer should be written in the answer booklet(s) provided.
5. All materials must be returned to the exam invigilator upon completion of the exam. Failure to do so will be deemed academic misconduct and will be dealt with accordingly.

Question 1. [3+8+14=25 MARKS]

A) Compute the following expressions that are given in postfix notation. Show the procedure.

5 7 * 3 4 1 + * +

[3 marks]

B) What is the asymptotic upper value of the expression $\sum_{i=1}^n i^2$ as a function of n by using the Big-O notation? Justify your answer.

[8 marks]

C) Give an upper tight bound of the runtime complexity class for each of the following two code fragments in Big-O notation, in terms of the variable N . Justify your answers.

1)

[7 marks]

```
int sum = 0;
for (int i = 1; i <= N - 5; i++) {
    for (int j = 1; j <= N - 5; j = j * 2) {
        sum++;
    }
}
```

2)

[7 marks]

```
int sum = N;
for (int i = 0; i < 1000; i++) {
    for (int j = 1; j <= i; j++) {
        sum += N;
    }
    for (int j = 1; j <= i; j++) {
        sum += N;
    }
    for (int j = 1; j <= i; j++) {
        sum += N;
    }
}
```

Question 2. [9 MARKS]

Given two 64 bit integers a and n , here is an algorithm to compute a^n

```
Power(a,n):
1. If  $n = 0$ : return 1.
2. If  $n = 1$ : return  $a$ .
3. If  $n$  is even:
4. Return  $\text{Power}(a \times a, n/2)$ .
5. else:
6. Return  $a \times \text{Power}(a \times a, (n-1)/2)$ .
```

Assume throughout this problem that we do not need to worry about overflow (a^n fits into a 64-bit integer variable) and that each operation on a 64-bit integer takes $O(1)$ time.

1) Let $A(n)$ denote the running time of $\text{Power}(a,n)$. Write a recurrence relation for $A(n)$.

[3 marks]

2) What is the solution (in Big-Theta notation) to your recurrence $A(n)$? Justify your answer.

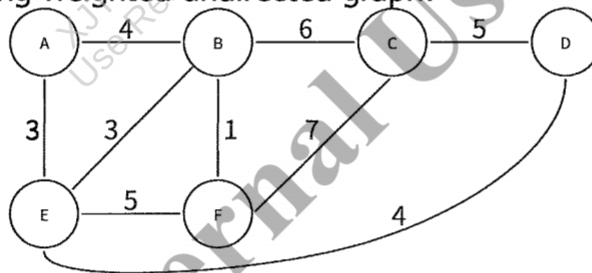
[6 marks]

Question 3. [16 MARKS]

Insert the following sequence of elements into an AVL tree, starting with an empty tree: 10, 20, 15, 25, 30, 16, 18, 19. Draw the AVL tree after each insertion. [16 marks]

Question 4. [10+5=15 MARKS]

Given the following weighted undirected graph:



(1) Show the step by step process of using the Prim's algorithm to find the minimum spanning tree. We start with node A. And draw the minimum spanning tree. [10 marks]

(2) What is the total weight of the minimum spanning tree? [5 marks]

Question 5. [8+5+7=20 MARKS]

Alice and Bob are using the RSA cryptosystem for secure communication. Bob has chosen two prime numbers $p=5$ and $q=17$ to generate his public and private keys.

(1) Compute n , $\Phi(n)$, and select a valid encryption key e (e is less than 10). [8 marks]

(2) Encrypt the message 4 according to the selected encryption key e . [5 marks]

(3) Derive the private key d . [7 marks]

Question 6. [5+10=15 MARKS]

Let $G=(V,E)$ be a directed graph. Let S be a subset of edges of G such that deletion of S (edges) results in a graph G' with no directed cycles. Such an S is a feedback arc set. The size of S is the number of edges in S . The Feedback Arc Set (FAS) decision problem is "to determine for a given integer k if G has a feedback arc set of size at most k ".

(1) Is the Feedback Arc Set (FAS) problem in NP? Prove or disprove it. **[5 marks]**

(2) Provide a reduction from the Vertex Cover (VC) problem to the Feedback Arc Set (FAS) problem, and prove its correctness. **[10 marks]**

END OF EXAM PAPER

Internal Use Only